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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
interface DeliveryCharge{
    public void calculateDeliveryCharge();
}
interface CouponDiscount{
    public void calculateDiscount();
}
public class FoodOrder implements DeliveryCharge,CouponDiscount{
    public double delivery_charge;
    public String coupon_code;
    public double discount;
    public double food_cost;
    public double final_bill;
    public void calculateDeliveryCharge(){
        if(food_cost>=500){
            delivery_charge=0;
        }
        else{
            delivery_charge=50;
        }
    }
    public void calculateDiscount(){
        if(coupon_code.equals("SAVE10")){
            discount=food_cost*0.1;
        }
        else{
            discount=0;
        }
    }
    public void calculateFinalBill(){
        final_bill=food_cost+delivery_charge-discount;
    }
    public void display(){
        System.out.println("Food Cost : " +food_cost);
        System.out.println("Delivery Charge : "+delivery_charge);
        System.out.println("Discount : "+discount);
        System.out.println("Final Bill : "+final_bill);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        FoodOrder f=new FoodOrder();
        f.food_cost=sc.nextDouble();
        sc.nextLine();
        f.coupon_code=sc.nextLine();
        f.calculateDeliveryCharge();
        f.calculateDiscount();
        f.calculateFinalBill();
        f.display();
    }
}
```

INPUT

600
SAVE10

OUTPUT

Food Cost: 600.0
Delivery Charge: 0.0
Discount: 60.0
Final Bill: 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        HotelBill h=new HotelBill();
        h.Room_number=sc.nextInt();
        sc.nextLine();
        h.Name=sc.nextLine();
        h.Rent=sc.nextDouble();
        h.calculateServiceCharge();
        h.finalbill();
        h.display();
    }
}
class Room{
    public int Room_number;
    public String Name;
    public double Rent;
}
interface Service_Charge{
    public double calculateServiceCharge();
}
class HotelBill extends Room implements Service_Charge{
    public double final_bill;
    public double ServiceCharge;
    public double calculateServiceCharge(){
        if(Rent>=5000){
            ServiceCharge=Rent*0.12;
        }
        else{
            ServiceCharge=Rent*0.08;
        }
        return ServiceCharge;
    }
    public void finalbill(){
        final_bill=ServiceCharge+Rent;
    }
    void display(){
        System.out.println("Room Number : "+Room_number);
        System.out.println("Guest Name : "+Name);
        System.out.println("Room Rent : "+Rent);
        System.out.println("Service Charge : "+ServiceCharge);
        System.out.println("Final Bill : "+final_bill);
    }
}
```

INPUT

205
Ananya
6000

OUTPUT

Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0